PERIODIC PROPERTIES 1

10

Properties or characteristics of elements change in an ordered manner moving across the Periodic Table.

Ionisation energy

Scientists obtained information about how electrons are arranged in atoms by studying how much energy is required to remove electrons from these atoms. When electrons (negative charge) are removed from atoms (neutral charge), positive ions are formed. Figure 1 shows how a gaseous lithium atom forms a positive ion when an electron is removed.

The term **first ionisation energy of an element** refers to the energy required to remove one electron from each atom in a mole of gaseous atoms.

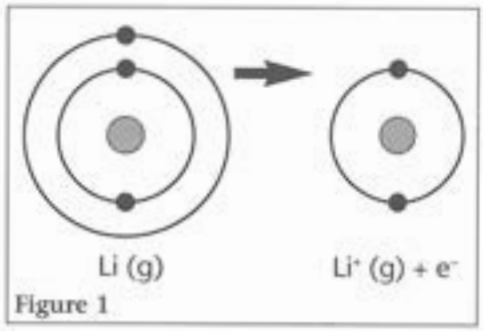


Figure 1

1 Why are positive ions formed when electrons are removed from electrically-neutral atoms?

Figure 2 shows a graph of ionisation energy (y-axis) against atomic number (x-axis) for the first 20 elements in the Periodic Table.

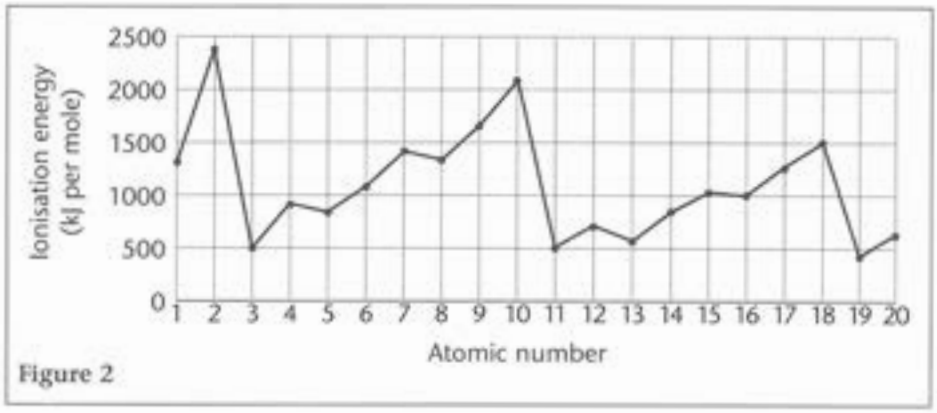


Figure 2

2 Give the name and symbol of each element.

- **a** atomic number 2
- **b** atomic number 10
- **c** atomic number 18

3 In which Group is each element in question 2?

4 Give the name and symbol of each element.

- **a** atomic number 3
- **b** atomic number 11
- **c** atomic number 19

5 In which Group is each element in question 4?

6 Give the name and symbol of each element.

- **a** atomic number 9
- **b** atomic number 17

7 In which Group is each element in question 6?

8 Which element in figure 2 has the highest ionisation energy?

9 Which element figure 2 has the lowest ionisation energy?

10 Describe what happens to the ionisation energies of elements as you move across a Period.

11 Describe what happens to the ionisation energies of elements as you move down a Group.

The large ionisation energies of the noble gas (Group VIII) elements indicate that they are unreactive and have stable electronic structures (i.e. full outer shells, or octets).

12 What do you think a low ionisation energy indicates about a particular element?

13 Mark each statement as true or false.

- **a** It takes energy to remove an electron from an atom.
- **b** Noble gas atoms lose electrons easily.
- **c** Electrons have a negative electrical charge.
- **d** There are eight electrons in the first shell of an atom.
- **e** Ionisation energies are an example of a Periodic property
- **f** Removing electrons from an atom produces a negative ion.
- **g** Ionisation energy is the energy used to remove electrons from gaseous atoms.
- **h** An atom's ionisation energy can be used to suggest properties which that atom may possess.

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